



RIVER VALLEY HIGH SCHOOL

YEAR 6 PRELIMINARY EXAMINATION

CANDIDATE
NAME

CENTRE
NUMBER

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INDEX
NUMBER

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CLASS

H1 BIOLOGY

8875/01

Paper 1 Multiple Choice

21 September 2011

1 hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, Centre number and index number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **thirty** questions in this paper. Answer **all** questions. For each question, there are four possible answers, **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate **Answer Sheet**.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

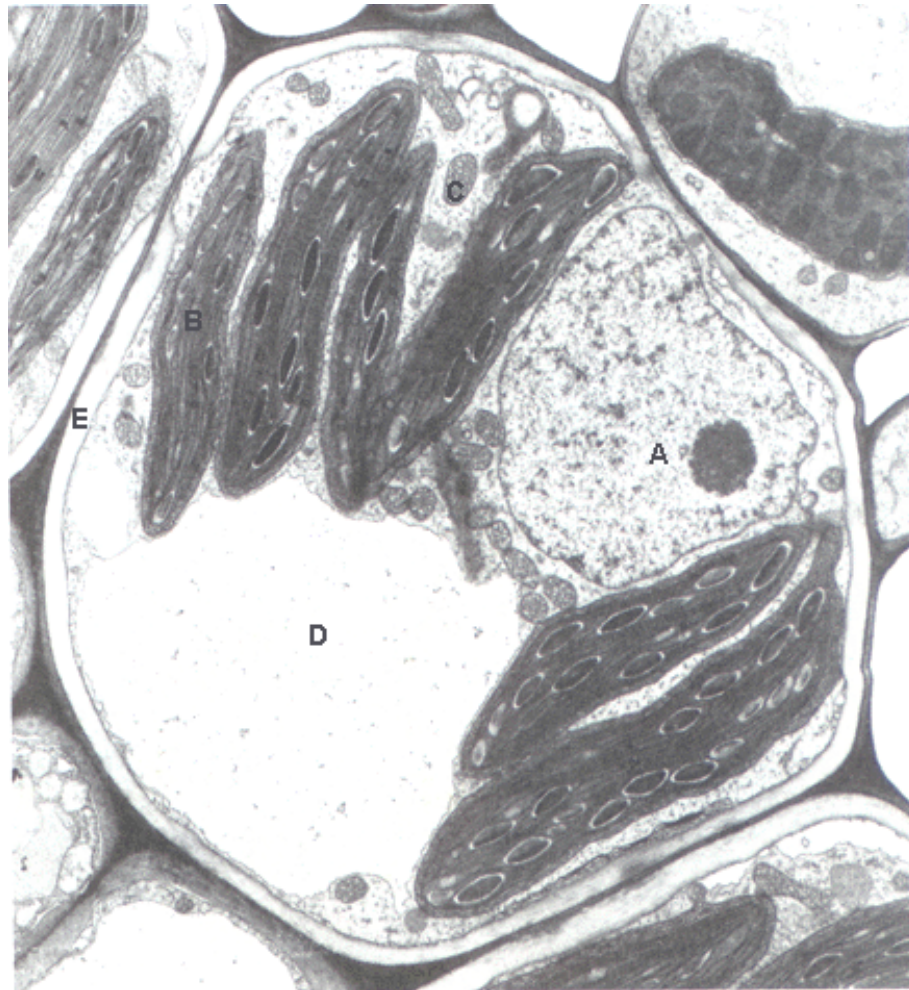
Any rough working should be done on the Question Paper.

This Question Paper consists of **18** printed pages.

For each question, there are four possible answers, **A**, **B**, **C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

1. **Fig. 1** represents part of a plant cell.



Source: <http://library.thinkquest.org/>

Fig. 1

If the cell has been growing in an environment in which its sole source of nitrogen had been labelled radioactively, we would expect to find radioactivity in structure(s)

- A** **A** only.
- B** **A**, **B** and **C** only.
- C** **B**, **C** and **D** only.
- D** **C**, **D** and **E** only.

2. **Fig. 2** is a diagram of the secretory pathway from the endoplasmic reticulum (ER), through the Golgi apparatus, to the plasma membrane.

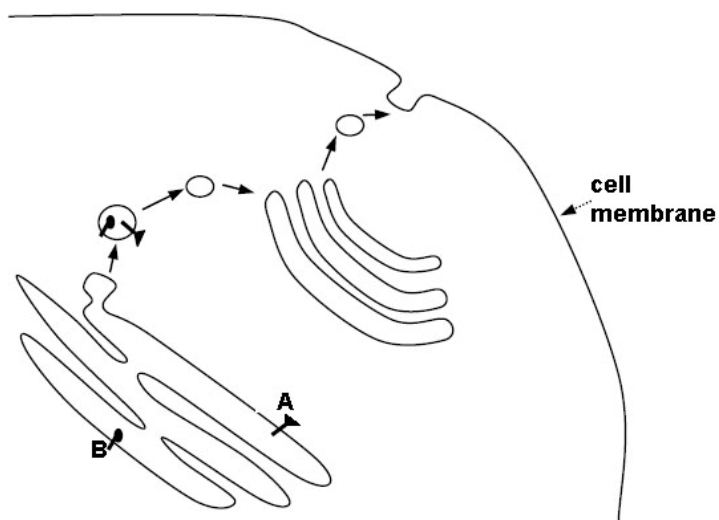
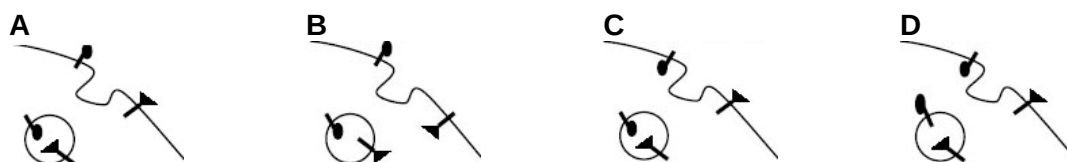
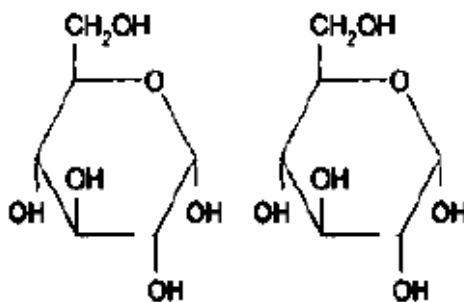


Fig. 2

Which of the following diagrams correctly represents the orientations of proteins **A** and **B** upon reaching the secretory vesicle and the plasma membrane?



3. The diagram shows two molecules.



What is formed when a large number of these molecules are condensed?

- A amylose
- B cellulose
- C phospholipid
- D protein

4. The four acids shown in **Fig. 4** form part of an enzyme-catalysed sequence.

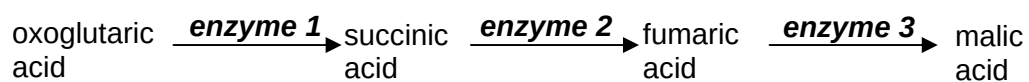


Fig. 4

The addition of malonic acid results in no change in the concentration of oxoglutaric acid, an accumulation of succinic acid, and a near absence of both fumaric acid and malic acid.

Further addition of fumaric acid results in the formation of malic acid.

What does this information indicate about malonic acid?

- A** It is an inhibitor of **enzyme 1**.
- B** It catalyses the formation of succinic acid.
- C** It is an inhibitor of **enzyme 2**.
- D** It reacts with fumaric acid.

5. **Fig. 5.1** is an electronmicrograph showing a cell at a particular stage during the cell cycle.

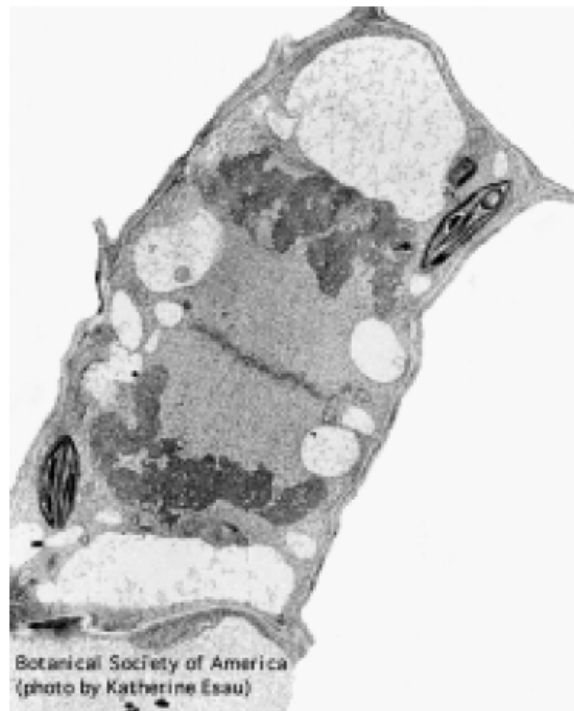


Fig. 5.1

The graph (**Fig. 5.2**) shows the mass of DNA within a cell during the cell cycle.

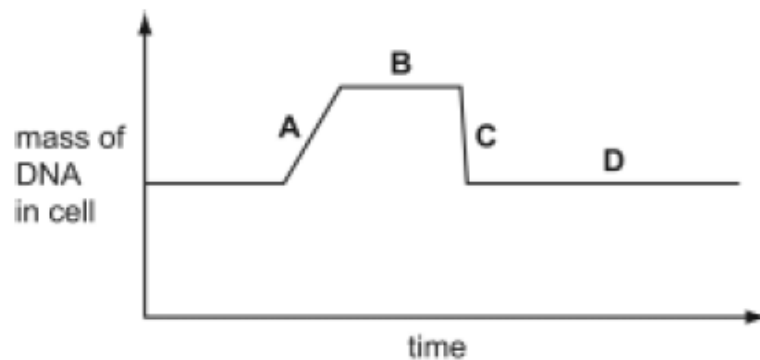


Fig. 5.2

When does the stage shown in **Fig. 5.1** occur?

6. A cell in the G1 phase has two homologous pairs of chromosomes. It undergoes a mitotic division, followed by meiosis. At the end of meiosis 2, what is the sum of all the products of these divisions in all cells?
- A 16 chromosomes, 16 chromatids, 8 cells, 4 times as many alleles.
- B 16 chromosomes, 16 chromatids, 4 cells, 4 times as many alleles.
- C 8 chromosomes, 16 chromatids, 8 cells, 2 times as many alleles.
- D 8 chromosomes, 16 chromatids, 16 cells, 4 times as many alleles.
7. As shown in **Fig. 7**, there are two bidirectional replication forks per replication bubble normally. If a mutant exists such that it has only one functional fork per replication bubble, which of the following would happen?

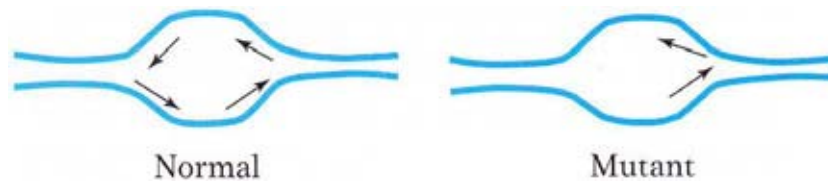


Fig. 7

- A DNA synthesis might take longer, but otherwise there would be no noticeable difference.
- B The DNA fragments from the two daughter strands could become mixed, producing possible mutations.
- C DNA synthesis would not take place because the appropriate enzymes to carry out discontinuous replication for both daughter strands would not be present.
- D DNA synthesis would not take place because the entire length of the chromosome would have to be unwound before both strands could be replicated in a discontinuous fashion.

8. Which of the following correctly matches the types of RNA molecules with their descriptions?

	<i>carries the information for the synthesis of one or more proteins</i>	<i>binds to an amino acid and guides it to the correct position on the ribosome</i>
A	transfer RNA	messenger RNA
B	ribosomal RNA	messenger RNA
C	messenger RNA	transfer RNA
D	messenger RNA	ribosomal RNA

9. Which of the following best describes the process of alternative splicing?

- A various ways of reading nucleotides in groups of three from a start codon
- B the succession of codons determined by reading nucleotides in groups of three from a specific start codon
- C the reaction that removes introns and joins together exons in eukaryotic nuclear primary RNA transcripts
- D various ways of splicing out introns in eukaryotic pre-mRNAs resulting in one gene producing several different mRNAs and protein products.

10. The diagram shows the first five codons of an mRNA molecule and the first five amino acids of the encoded polypeptide.

AUG	-	AAG	-	UGU	-	UAU	-	CUC	-	AUG
methionine		lysine		cysteine		tyrosine		leucine		methionine

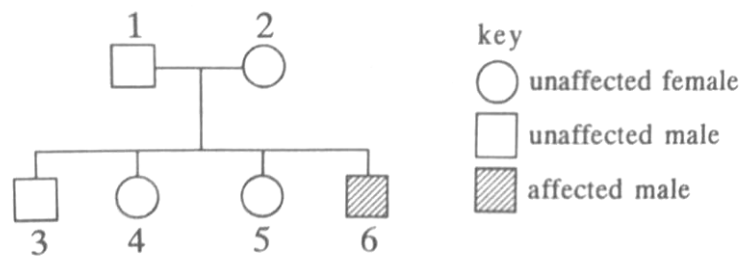
As a result of a mutation, a dipeptide was produced instead of the normal polypeptide.

The triplets on the DNA strand that is complementary to mRNA, that code for 'stop', are ATT, ATC and ACT.

Which statement explains the production of a dipeptide?

- A The DNA triplet coding for methionine is changed from TAC to ATC.
- B The DNA triplet coding for lysine is changed from TTC to ATC.
- C The DNA triplet coding for cysteine is changed from ACA to ACT.
- D The DNA triplet coding for tyrosine is changed form ATA to ATG

11. The diagram shows the inheritance of haemophilia in a family.



If daughter 4 married a normal male, what is the probability that their first child would suffer from haemophilia?

- A 0 B 1 in 8 C 1 in 4 D 1 in 2
12. In a species of plant, the allele for tall stems is dominant over that for short stems and the allele for purple flowers is dominant over that for white flowers.

A cross between two plants produced the following phenotypic ratio.

tall, purple-flowered	3
tall, white-flowered	3
short, purple-flowered	1
short, white-flowered	1

What are the phenotypes of the parents?

- | | | |
|---|------------------------|-----------------------|
| A | short, purple-flowered | tall, purple-flowered |
| B | short, white-flowered | tall, purple-flowered |
| C | tall, purple-flowered | tall, white-flowered |
| D | tall, white-flowered | short, white-flowered |

13. The following image is of a Siamese cat.



The colour produced in the fur of a Siamese cat is under the control of a tyrosinase enzyme that is sensitive to particular temperatures. A summary of the enzyme action is

pigment precursor $\xrightarrow{\text{body temperature at or above } 37^{\circ}\text{C}}$ no pigment produced

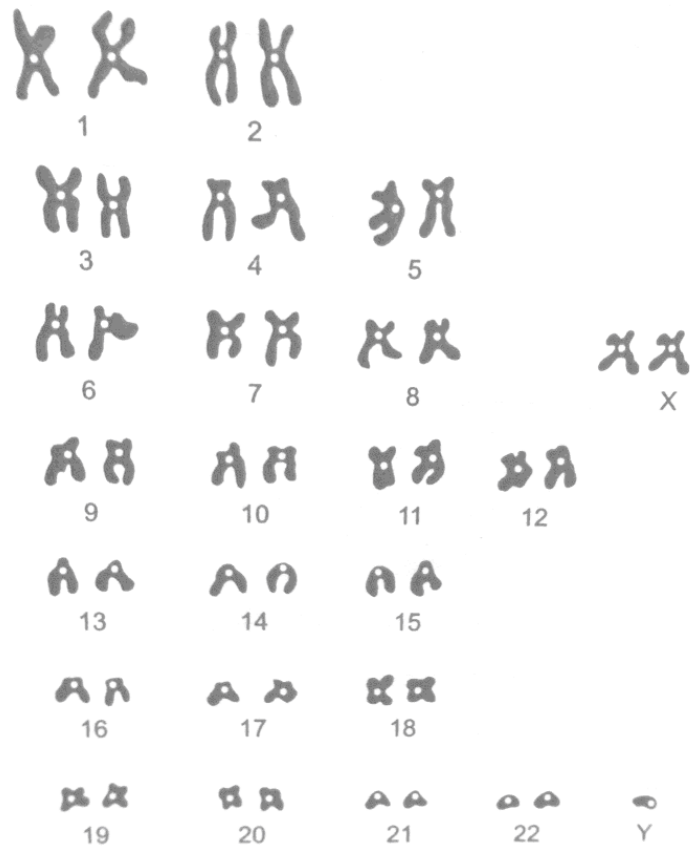
pigment precursor $\xrightarrow{\text{body temperature below } 37^{\circ}\text{C}}$ pigment produced

The core body temperature of a cat is around 38°C .

It is likely that in London

- A Siamese kittens have dark fur over their bodies when they are born.
- B Siamese cats that live outside a house show an increase in pigment during summer.
- C Siamese cats that live inside a house in winter have darker fur than those that live outside.
- D Siamese cats that live outside in winter have darker fur than Siamese cats in tropical regions.

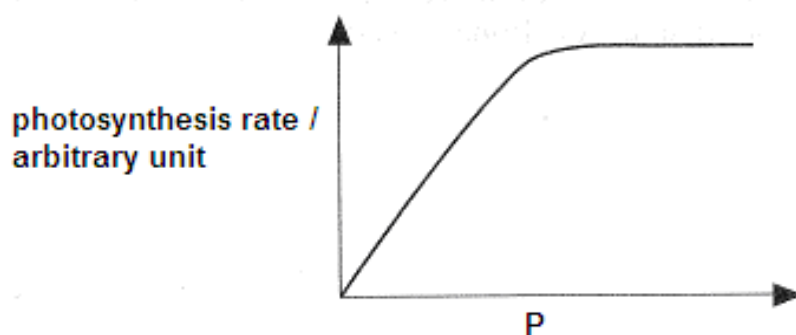
14. The diagram shows a human karyogram.



What can be concluded from the karyotype provided?

- A There was non-disjunction during meiosis I in the mother.
- B There was non-disjunction during meiosis II in the father.
- C One contributory gamete to the zygote is an egg with an X and a Y chromosome.
- D One contributory gamete to the zygote is a sperm containing two X chromosomes.

15. The following graph shows the relationship between the rates of photosynthesis with environmental factor **P**.



P most probably represents

- 1 carbon dioxide concentration
- 2 oxygen concentration
- 3 light intensity
- 4 temperature

A 1 and 2

B 1 and 3

C 1 and 4

D 1, 3 and 4

16. It has been found that an aqueous suspension of isolated chloroplasts will evolve oxygen if illuminated in the presence of a certain type of compound.

Which type of compound and which colours of light are required for maximum oxygen evolution?

	<i>type of compound</i>	<i>colours of light at which maximum evolution occurs</i>
A	electron acceptor	blue and green
B	electron acceptor	blue and red
C	electron donor	blue and green
D	electron donor	blue and red

17. The weedkiller DCMU blocks the flow of electrons from the electron transport chains in photophosphorylation.

Why does this kill the plant?

- A ATP and reduced NADP are not produced.
- B ATP and reduced NAD are not produced.
- C Photoactivation of the chlorophyll cannot occur.
- D Photolysis of water does not occur.

18. The reaction below occurs in all aerobic cells.

Pyruvate $\rightarrow$ Acetylcoenzyme A

Which of the following statements are true of the above reaction?

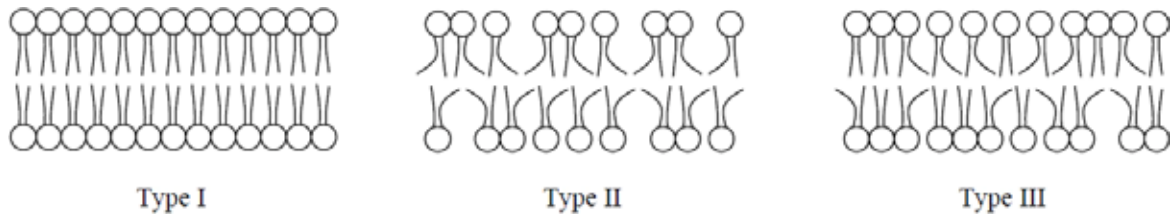
- 1 The reaction happens in the cytoplasm of the cells.
- 2 The above reaction involved oxidative decarboxylation.
- 3 2H are released and received by NAD^+ .
- 4 ATP energy is used in the above reaction.
- 5 Carbon dioxide is released during this reaction.

- A 1 and 3 B 2 and 4 C 1, 2 and 4 D 2, 3 and 5

19. Which row in the table describes the first stage of cellular respiration?

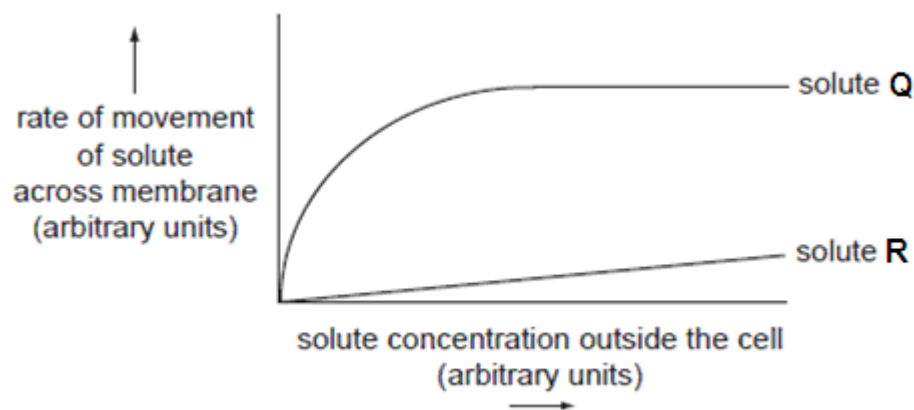
	<i>Substrate</i>	<i>Location</i>	<i>Product</i>	<i>Product</i>
A	pyruvate	mitochondria	oxygen	water
B	pyruvate	cytoplasm	carbon dioxide	ATP
C	glucose	mitochondria	pyruvate	water
D	glucose	cytoplasm	pyruvate	ATP

20. A mutated strain of the bacterium *Escherichia coli* was found to be incapable of incorporating unsaturated phospholipids into its plasma membrane. Which of the following correctly depicts and describes the membrane of such a bacteria?



- A The membrane would appear as Type I and would be more fluid at low temperatures.
- B The membrane would appear as Type I and would be less fluid at low temperatures.
- C The membrane would appear as Type II and would be more fluid at low temperatures.
- D The membrane would appear as Type III and would be no different from normal bacterial membranes.
21. Solute **Q** moves across the plasma membrane via protein channels. Solute **R** moves across the plasma membrane by simple diffusion. The rate of movement of each solute into a cell is recorded and graphed.

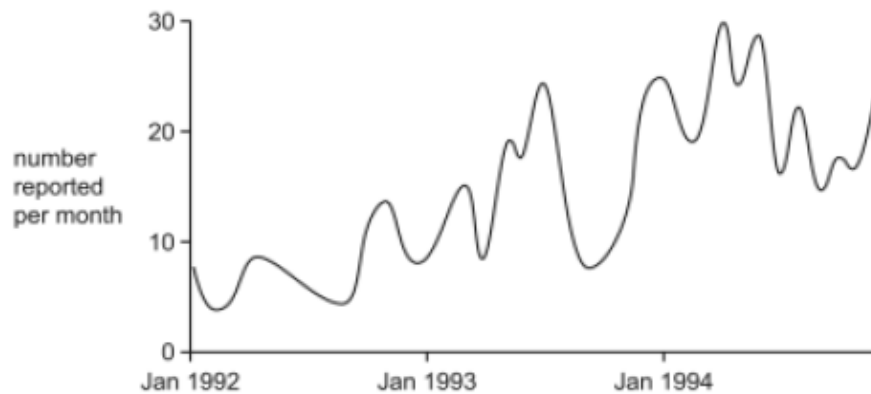
The results are shown in the following graph.



It would be reasonable to conclude that

- A solute **Q** is lipid soluble.
- B solute **Q** saturates the protein channels.
- C the gradient of the graph for solute **R** will increase as the temperature decreases.
- D as solute **R** is metabolised within a cell, its rate of movement across the membrane decreases.

22. The graph shows the change in number of bacterial samples from some New York hospitals that were resistant to the antibiotic vancomycin in 1992–4. 40 samples were taken each month from randomly selected patients who had become infected with bacteria in hospital.



Which of the following most accurately describes the cause of the changes in the frequency of the vancomycin resistant phenotype that occurred?

- A effect of artificial selection
 - B effect of natural selection
 - C purely due to genetic drift
 - D purely due to random sampling effects
23. The table shows the functions of three bones, **M**, **I** and **S**, in three classes of vertebrates. Each bone arises in the same way in embryos of the three classes.

Class	M	I	S
Fish	part of the articulation of upper and lower jaws	part of the articulation of upper and lower jaws	attaches the upper and lower jaws to the skull
Amphibian	part of the articulation of the lower jaw	part of the articulation of the lower jaw	transmits sound to the inner ear
Mammals	transmits sound to the inner ear	transmits sound to the inner ear	transmits sound to the inner ear

How do the functions of these bones support Darwin's theory of natural selection?

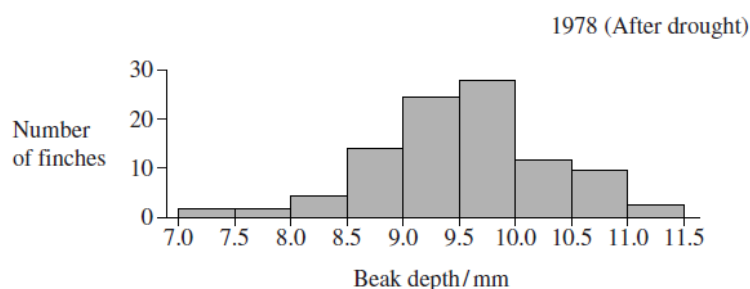
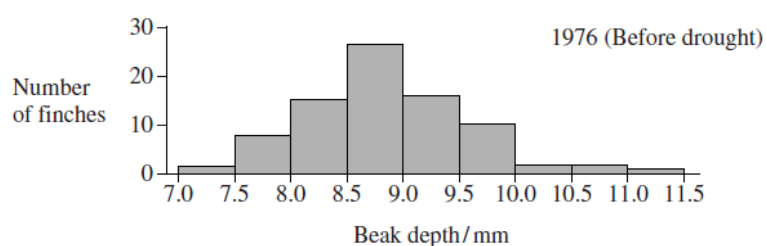
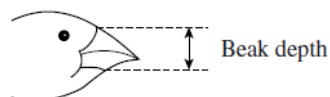
- A They show convergent evolution of analogous structures.
- B They show divergent evolution of analogous structures.
- C They show how analogous structures have become adapted to perform different functions.
- D They show how homologous structures have become adapted to perform different functions.

24. The ground finch, *Geospiza fortis*, is a species of bird which lives on a small isolated island.

These finches feed on seeds of different sizes from different species of plants. The finches show variation in the size of their beaks. Birds with larger beaks can eat large and small seeds. Birds with smaller beaks are only able to eat small seeds.

In 1977 there was a severe drought on the island. This killed many species of plants that the finches fed on. One species of food plant did survive and this produced large seeds.

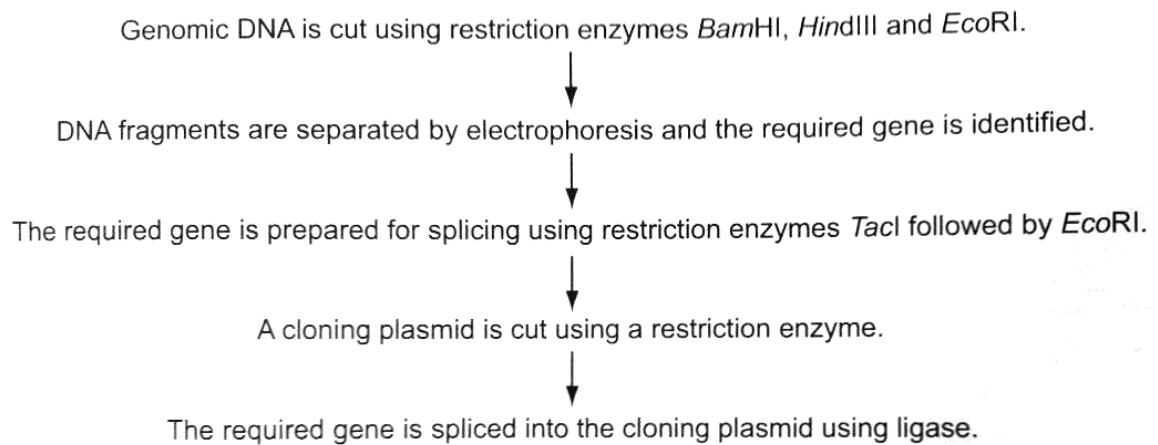
The graphs show the distribution of beak sizes of the finch population before and after the drought. Beak size was measured by the depth of the beak, as shown in the diagram.



Which type of selection and inheritance does this show?

	<i>type of selection</i>	<i>inheritance</i>
A	Directional	Polygenic
B	Directional	Multiple alleles
C	Stabilising	Polygenic
D	Stabilising	Multiple alleles

25. The flow diagram shows the synthesis of a recombinant DNA molecule.



Which restriction enzyme is used to cut the plasmid before the required gene is spliced?

- A *Bam*HI
- B *Eco*RI
- C *Hind*III
- D *Tac*I
26. Which primers should be paired to amplify, by PCR, the DNA flanked by the two indicated sequences?

5' – GGAATTCGT -- // -- AATGCTACC – 3'

- | | |
|-------------------------|-------------------------|
| A ACGAATTCC & AATCGTACC | B AATGCTACC & GGAATTCGT |
| C GGTAGCATT & GGAATTCGT | D ACGAATTCC & GGTAGCAAT |

27. A family exhibits a rare autosomal recessive disease, which is caused by a point mutation that abolishes an *Eco*R1 restriction site in the genome. To ascertain the genotype of family members, researchers (i) amplify a 1.8 kb region of this area using PCR; (ii) cut the PCR products with *Eco*R1; and then (iii) separate the DNA fragments on an agarose gel. The results of such an experiment are shown in **Fig. 27** below, using DNA obtained from each of the five family members.

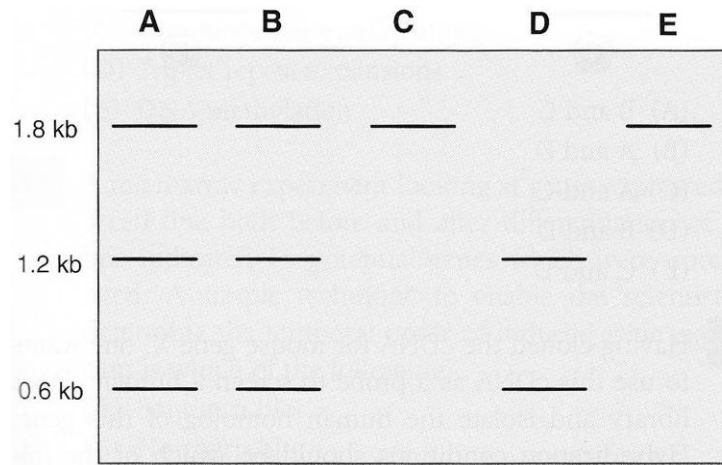


Fig. 27

Which family members have the disease?

- | | | | |
|----------|----------------|----------|-------------------|
| A | A and B | B | C and E |
| C | D only | D | A, B and D |

- 28.** Which of the following best explains why a genome project is not finished when the sequence has been completed?
- A** Genomes change too fast and must constantly be updated.
 - B** Without knowing the number, the function, and the location of genes within a genome, the sequence is not very useful.
 - C** Individuals within a species are so unique that having a single sequence is insufficient to characterise a species' genome.
 - D** Many sections of a genome are too difficult to sequence and have not actually been included in 'complete' genomes.

29. Which of these processes could not increase crop yield?

- P** inserting genes for vitamin A production into rice
- Q** inserting genes for pest resistance into cotton
- R** inserting genes for herbicide resistance into oilseed rape
- S** inserting genes for nitrogen fixation into soya beans

A **P** and **S**

B **Q** and **R**

C **R** and **S**

D **Q** and **P**

30. Embryonic stem cells are widely used in medical research.

Which of the following properties of stem cells makes them particularly useful in this research?

- 1** They are totipotent.
- 2** They can be made to differentiate into various types of cell.
- 3** They can divide indefinitely.
- 4** They are readily available than other cells.

A **1** and **3**

B **2** and **3**

C **1, 2** and **4**

D **2, 3** and **4**